Reinforcement Learning for Beer Game Inventory Control

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Abstract

This report studies single-agent reinforcement learning for inventory ordering in the Beer Game. The current codebase trains one learning firm in a three-firm serial supply chain while the other firms are controlled by simple base-stock policies. We compare three standard-action agents, DQN, Double DQN, and PPO, and three high-dimensional action agents, High-Dim DQN, High-Dim Double DQN, and High-Dim PPO. All reported results are taken from the latest complete runs in the current results/ directory and the refreshed report figures. In the standard action setting, Double DQN obtains the highest mean test reward, 631.92, while PPO has the lowest standard deviation, 39.30, and the highest demand satisfaction rate, 0.934, over 20 test episodes. In the high-dimensional setting, High-Dim Double DQN reaches the highest mean reward, 445.50, while the newly added High-Dim PPO run reaches 406.50 with lower variance than the high-dimensional value-based agents. The results suggest that PPO remains the most stable standard-action policy, while the vector-order setting is sensitive to both algorithm choice and optimization details.

1. Introduction

The Beer Game is a classical supply-chain decision problem in which firms place orders along a serial chain while facing stochastic demand, inventory holding cost, and shortage penalties. A locally reasonable order decision may still create excessive inventory, unsatisfied demand, and amplified supply-chain fluctuation. This makes the task a useful benchmark for reinforcement learning methods that must balance short-term service quality against long-term inventory cost.

The project originally implemented a DQN baseline. The current experiment pipeline has since been extended in four directions. First, Double DQN is included as a value-based improvement that decouples action selection from target evaluation. Second, PPO is added as an on-policy actor-critic method with clipped policy updates. Third, the scalar order action is extended to a high-dimensional order vector, which can represent multi-channel or multi-supplier order allocation. Fourth, PPO is also applied to this high-dimensional action space by treating the enumerated vector orders as categorical policy actions.

This report updates the previous project write-up using the current experiment results. It focuses on the actual runs saved under results/, rather than earlier figures or obsolete command formats. The main questions are:

- Which standard-action algorithm gives the best reward and stability?
- Does Double DQN improve over DQN in the present setting?
- How much harder is the high-dimensional action formulation, and does PPO help in that setting?
- What should be adjusted in future training and experiment management?

2. Environment and Objective

2.1. Beer Game Model

The environment contains $N = 3$ firms in a serial supply chain. Firm 0 is closest to the external market, firm 1 is the learning firm in all experiments, and firm 2 is the upstream supplier. Each episode lasts at most 100 time steps. External market demand follows a Poisson distribution:

$$D_t \sim \text{Poisson}(10). \tag{1}$$

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For the standard-action agents, the learning firm chooses one discrete order quantity:

$$a_t \in \{1, 2, \ldots, 20\}. \quad (2)$$

The local observation contains the firm's order, satisfied demand, and inventory level. Inventory evolves according to the standard Beer Game balance:

$$I_{t+1} = I_t + q_t - d_t, \quad (3)$$

where q_t is the received or ordered quantity represented by the environment transition and d_t is satisfied demand.

2.2. Reward

The learning objective is to maximize cumulative firm profit. The instantaneous reward combines sales revenue, procurement cost, holding cost, and shortage penalty:

$$r_t = p_i d_t - p_{i+1} q_t - h I_t - c \max(0, D_t - d_t). \quad (4)$$

The experiments use prices $[10, 9, 8]$, holding cost $h = 0.5$, shortage penalty $c = 2.0$, initial inventory 100, and a maximum episode length of 100.

2.3. Non-Learning Firms

The non-learning firms use a base-stock heuristic. This keeps the environment more stable than random opponents and makes algorithm comparison easier to interpret. The base-stock policy orders according to the gap between a target inventory and current inventory, clipped to the feasible action range.

3. Algorithms

3.1. DQN

DQN approximates $Q(s, a)$ with a neural network and trains from replay-buffer samples. The target is

$$y = r + \gamma \max_{a'} Q_{\text{target}}(s', a'). \quad (5)$$

The current DQN run uses 2000 training episodes, 20 test episodes, learning rate 5×10^{-4}, discount factor $\gamma = 0.99$, replay buffer size 10000, batch size 64, and soft target-network update coefficient 10^{-3}.

3.2. Double DQN

Double DQN reduces the maximization bias of standard DQN by selecting the next action with the online network and evaluating it with the target network:

$$a^* = \arg\max_{a'} Q_{\text{online}}(s', a'), \quad (6)$$

$$y = r + \gamma Q_{\text{target}}(s', a^*). \quad (7)$$

The current Double DQN run uses the same DQN hyperparameters as the standard DQN comparison run, except for the seed offset used to separate random streams.

3.3. PPO

PPO trains a stochastic policy and value function with clipped policy-gradient updates. The current run uses learning rate 10^{-4}, discount factor 0.99, GAE parameter 0.95, clip range 0.15, entropy coefficient 0.03, four update epochs, minibatch size 64, and reward scaling 0.01. PPO is included because the Beer Game ordering policy can benefit from a directly optimized stochastic policy, especially when the value-function estimates of DQN-style methods are noisy.

3.4. High-Dimensional Action Agents

The high-dimensional setting replaces the scalar order with a three-dimensional order vector. Each dimension can be interpreted as a channel or supplier. The available per-channel levels are $\{0, 5, 10, 15, 20\}$, and the total order is constrained to be between 5 and 20. After filtering invalid combinations, the action set contains 34 discrete composite actions. DQN and Double DQN learn Q-values over these composite actions. High-Dim PPO uses the same enumeration, but trains a categorical actor-critic policy over the 34 vector-order choices and maps the selected category back to an order vector before stepping the environment.

4. Experiment Management

The current experiments are managed by Hydra configuration files in cfg/. Each training script writes its own outputs to a timestamped directory under results/<algorithm>/. A run directory contains the resolved Hydra configuration, logs, model checkpoint, training scores, test scores, test trajectory history, runtime metadata, and summary.json. The plotting script then reads selected run directories and writes consolidated figures under results/summary/<timestamp>/.

5. Results

5.1. Standard-Action Results

?? summarizes the latest standard-action runs. Double DQN obtains the highest mean reward, but its advantage over DQN is small: 0.85 reward points. PPO has a slightly lower mean reward than Double DQN, but it has the lowest test-reward standard deviation and the highest demand satisfaction rate. Thus the

Table 1. Latest complete runs used in the main comparison.

Algorithm	Run directory	Train eps.	Runtime
DQN	20260629_200133	2000	173.02
Double DQN	20260629_200430	2000	171.98
PPO	20260629_200726	2000	164.37
High-Dim DQN	20260629_201014	2000	188.58
High-Dim Double DQN	20260629_201327	2000	175.79
High-Dim PPO	20260629_193915	2000	162.55

Table 2. Standard-action test performance over 20 test episodes.

Metric	DQN	Double DQN	PPO
Mean reward	631.08	631.92	627.90
Reward std.	67.19	48.77	39.30
Avg. inventory	16.58	16.37	17.41
Avg. order	8.37	8.18	8.45
Satisfied demand	9.26	9.07	9.32
Demand satisfaction	0.914	0.917	0.934

current standard-action comparison does not have a single dominant winner: Double DQN is marginally best in mean reward, while PPO is the most stable and service-oriented policy.

5.2. Training Dynamics

?? shows the moving-average training reward. PPO improves fastest and avoids the very large negative early returns observed in the high-dimensional value-based agents. DQN and Double DQN both recover from poor early exploration and converge to a similar reward range. The high-dimensional agents start with much lower rewards and recover more slowly, which is consistent with their larger composite action space. The new High-Dim PPO run follows the same difficult early phase but later reaches a higher test reward than the high-dimensional value-based agents.

5.3. Inventory and Ordering Behavior

?? compares average inventory, order quantity, and unsatisfied demand during testing. All methods draw down the initial inventory quickly and then operate near a lower steady inventory band. Among the scalar-action agents, PPO has the highest demand satisfaction rate, while Double DQN uses the lowest average inventory and order quantity. The high-dimensional agents keep higher inventory on average but still do not fully convert this into scalar-action-level reward, indicating that the extra action dimensions introduce cost and exploration difficulty. However, High-Dim PPO has lower reward variance than the high-dimensional value-based agents, and High-Dim Double DQN achieves the best high-dimensional mean reward in the refreshed comparison.

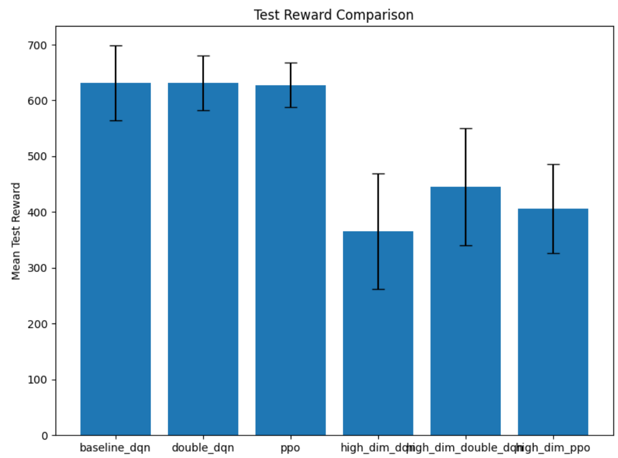

Figure 1. Mean test reward comparison. Error bars show one standard deviation across 20 test episodes.

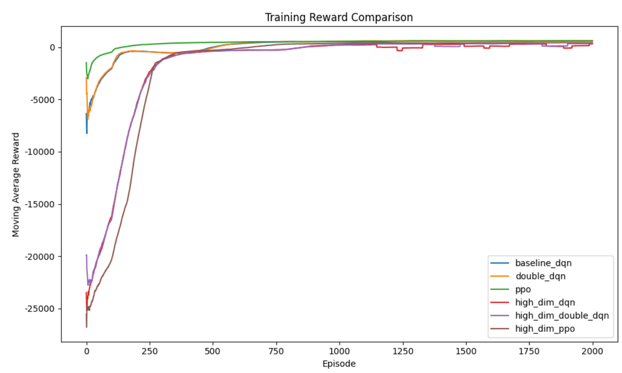

Figure 2. Training reward moving average for the latest complete runs.

5.4. High-Dimensional Results

?? reports the latest high-dimensional runs with the same 2000-episode training budget. High-Dim Double DQN obtains the highest mean reward among the high-dimensional agents, reaching 445.50. High-Dim PPO reaches 406.50, which is higher than High-Dim DQN and has the lowest reward standard deviation among the high-dimensional methods. High-Dim Double DQN has the highest demand satisfaction rate, 0.944. Nevertheless, even the best high-dimensional reward remains below the scalar-action agents, show-

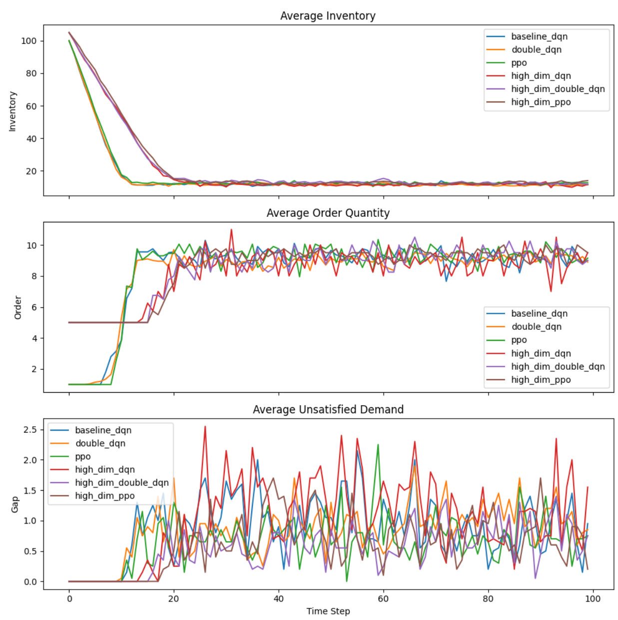

Figure 3. Average inventory, order quantity, and unsatisfied demand during testing.

Table 3. High-dimensional action test performance over 20 test episodes.

Metric	DQN	Double DQN	PPO
Mean reward	365.32	445.50	406.50
Reward std.	103.79	105.25	79.60
Avg. inventory	20.61	21.69	21.92
Avg. order	8.24	8.46	8.51
Satisfied demand	9.13	9.34	9.37
Demand satisfaction	0.901	0.944	0.939

ing that the vector-order formulation remains harder under the current enumerated action design.

6. Discussion

The main current conclusion is that the scalar-action methods still dominate the high-dimensional methods. Double DQN has the highest scalar mean reward in the refreshed comparison, while PPO has the lowest variance and highest scalar demand satisfaction rate. PPO's advantage is not caused by much higher average ordering: its average order quantity remains close to DQN and Double DQN, suggesting a better balance between inventory protection and holding cost.

Double DQN marginally beats DQN in mean reward in the current scalar setting and substantially reduces test-reward variance from 67.19 to 48.77. It also uses lower average inventory and order quantity, so its value is mostly risk reduction and leaner inventory behavior rather than a large reward improvement.

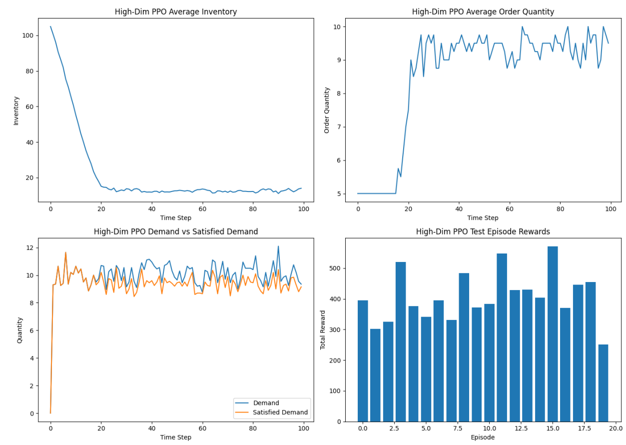

Figure 4. High-Dim PPO inventory, ordering, demand satisfaction, and test rewards.

Table 4. Additional high-dimensional exploratory runs currently present in results/.

Algorithm	Run	Train eps.	LR	Mean reward
High-Dim DQN	162220	1500	10^{-3}	351.70
High-Dim Double DQN	163325	1500	10^{-3}	406.10

The high-dimensional formulation is still not competitive with scalar-action agents. Although it can represent richer order allocation, the composite action space makes exploration harder and increases value-estimation or policy-search variance. High-Dim Double DQN improves the best high-dimensional mean reward to 445.50 in the refreshed comparison. High-Dim PPO does not beat that mean reward, but it has the lowest high-dimensional reward variance, suggesting that policy-gradient training is still a useful direction. The gap to scalar methods remains large, so the current enumerated vector-action design has not yet produced a better ordering policy than the simpler scalar formulation.

7. Recommendations

Future training should prioritize three changes. First, repeat each algorithm over multiple seeds and report confidence intervals; the current results are single-run comparisons and should not be interpreted as statistically final. Second, keep PPO as the main standard-action baseline and tune it around entropy coefficient, clip range, and reward scaling. Third, continue the high-dimensional PPO direction, but redesign the policy to avoid treating every composite order as an unrelated categorical action. A factorized policy or autoregressive order allocator would better match the multichannel structure and reduce exploration difficulty.

For experiment management, the current Hydra-based layout should be retained. Each run already saves runtime metadata, resolved configuration, logs, model files, scores, and summaries under timestamped results/<algorithm>/ directories. The report should continue to state exactly which run directories are used, because the plotting script selects the latest complete run by algorithm name unless explicit result paths are supplied.

8. Conclusion

Using the current saved results, scalar-action agents remain the strongest policies for the Beer Game inventory-control task. Double DQN obtains the best scalar mean reward, 631.92, while PPO has the lowest scalar reward standard deviation, 39.30, and the highest scalar demand satisfaction rate, 0.934. In the high-dimensional action setting, High-Dim Double DQN obtains the best mean reward, 445.50, and High-Dim PPO achieves a competitive mean reward of 406.50 with lower variance than the high-dimensional value-based methods. However, high-dimensional policies are still below scalar policies, so the next experimental step should be multi-seed evaluation and targeted tuning of factorized or autoregressive high-dimensional action policies.